MA1522 Linear Algebra for Computing
L2 Lesson 18



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Semester 1, 2026/27

Probability Vector and Stochastic Matrix

A vector $\boldsymbol{v} = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$ is a **probability vector** if and only if $v_1, v_2, \ldots, v_n \geq 0$ and $v_1 + v_2 + \cdots + v_n = 1$.

A square matrix $\boldsymbol{A}$ of order n is a **stochastic matrix** if and only if all its columns are probability vectors, i.e. if $\boldsymbol{A} = (a_{ij})$ then $a_{ij} \geq 0$ for all i, j and $a_{1j} + a_{2j} + \cdots + a_{nj} = 1$ for all j.

Note

Let $\boldsymbol{A}$ be a stochastic matrix of order n.

1. $\begin{pmatrix} 1 & 1 & \cdots & 1 \end{pmatrix} \boldsymbol{A} = \begin{pmatrix} 1 & 1 & \cdots & 1 \end{pmatrix}$.
2. If $\boldsymbol{v} \in \mathbb{R}^n$ is a probability vector, then $\boldsymbol{Av}$ is a probability vector.

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Question 1

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Let $\boldsymbol{A}$ be a stochastic matrix of order n. Which of the following statements is/are true?

a. 1 is an eigenvalue of $\boldsymbol{A}$.

b. If $\boldsymbol{v} = (v_1, v_2, \ldots, v_n)^T$ is an eigenvector of $\boldsymbol{A}$ associated with eigenvalue $\lambda \neq 1$, then $v_1 + v_2 + \cdots + v_n = 0$.



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Thus 1 is an eigenvalue of $\boldsymbol{A}^T$, and hence of $\boldsymbol{A}$.

b. If $\boldsymbol{v} = (v_1, v_2, \ldots, v_n)^T$ is an eigenvector of $\boldsymbol{A}$ associated with eigenvalue $\lambda \neq 1$, then $v_1 + v_2 + \cdots + v_n = 0$.

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Thus $(1 - \lambda)(v_1 + v_2 + \cdots + v_n) = 0$. Since $1 - \lambda = 0$, we must have $v_1 + v_2 + \cdots + v_n = 0$.

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Markov Chains

A **Markov chain** is an infinite sequence $(\boldsymbol{x}_k)_{k\in\mathbb{Z}_{\geq 0}}$ of probability vectors in $\mathbb{R}^n$ such that for some stochastic matrix $\boldsymbol{A}$, $\boldsymbol{x}_{k+1} = \boldsymbol{A}\boldsymbol{x}_k$ for all $k \in \mathbb{Z}_{\geq 0}$.

Equivalently, $\boldsymbol{x}_k = \boldsymbol{A}^k\boldsymbol{x}_0$ for all $k \in \mathbb{Z}^+$.

If $\boldsymbol{A}$ is diagonalizable, with $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$, then

$$\boldsymbol{A}^k = \boldsymbol{P}\boldsymbol{D}^k\boldsymbol{P}^{-1},$$

for all $n \in \mathbb{Z}^+$, and $\boldsymbol{D}^k = \operatorname{diag}(d_1^k, d_2^k, \ldots, d_n^k)$ if $\boldsymbol{D} = \operatorname{diag}(d_1, d_2, \ldots, d_n)$.

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Question 2

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A population of ants is put into a maze with 3 compartments A, B and C.

- If an ant is in compartment A, an hour later, there is a 20% chance it will go to compartment B, and a 40% chance it will go to compartment C.
- If it is in compartment B, an hour later, there is a 10% chance that it will go to compartment A, and a 30% chance it will go to compartment C.
- If it is in compartment C, an hour later, there is a 50% chance that it will go to compartment A, and a 20% chance that it will go to compartment B.

Suppose that 1000 ants have been placed in compartment A. Let $\boldsymbol{x}_n = \begin{pmatrix} a_n \\ b_n \\ c_n \end{pmatrix}$, where a_n, b_n, c_n are the number of ants in compartments A, B, C, respectively, after n hours.

a. Find a matrix $\boldsymbol{A}$ such that $\boldsymbol{x}_{n+1} = \boldsymbol{A}\boldsymbol{x}_n$.

b. Diagonalize $\boldsymbol{A}$ and estimate the number of ants in each compartment after 3 hours.

c. In the long run (assuming no ants died), estimate the number of ants in each compartment.

Question 2: Solution

a) $$\begin{aligned} a_{n+1} &= 0.4a_n + 0.1b_n + 0.5c_n. \\ b_{n+1} &= 0.2a_n + 0.6b_n + 0.2c_n. \\ c_{n+1} &= 0.4a_n + 0.3b_n + 0.3c_n. \end{aligned}$$ Thus, $\boldsymbol{A} = \begin{pmatrix} 0.4 & 0.1 & 0.5 \\ 0.2 & 0.6 & 0.2 \\ 0.4 & 0.3 & 0.3 \end{pmatrix}$.

b) $\det(x\boldsymbol{I} - \boldsymbol{A}) = (x-1)(x-\frac{2}{5})(x+\frac{1}{10})$.

$E_1 = \text{span}\left\{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right\}$, $E_{\frac{2}{5}} = \text{span}\left\{\begin{pmatrix} 4 \\ -5 \\ 1 \end{pmatrix}\right\}$, $E_{-\frac{1}{10}} = \text{span}\left\{\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}\right\}$.

Let $\boldsymbol{P} = \begin{pmatrix} 1 & 4 & -1 \\ 1 & -5 & 0 \\ 1 & 1 & 1 \end{pmatrix}$, $\boldsymbol{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{2}{5} & 0 \\ 0 & 0 & -\frac{1}{10} \end{pmatrix}$. Then $\boldsymbol{P}^{-1} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{15} & -\frac{2}{15} & \frac{1}{15} \\ -\frac{2}{5} & -\frac{1}{5} & \frac{3}{5} \end{pmatrix}$ and $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$.

So $\boldsymbol{x}_3 = \boldsymbol{A}^3\boldsymbol{x}_0 = \boldsymbol{P}\boldsymbol{D}^3\boldsymbol{P}^{-1}\begin{pmatrix} 1000 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 350 \\ 312 \\ 338 \end{pmatrix}$, i.e. there will be 350 ants in A, 312 in B and 338 in C after 3 hours.

c) As $n \to \infty$, $\boldsymbol{D}^n = \text{diag}(1^n, \left(\frac{2}{5}\right)^n, (-\frac{1}{10})^n) \to \text{diag}(1, 0, 0)$,

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So $\boldsymbol{x}_3 = \boldsymbol{A}^3\boldsymbol{x}_0 = \boldsymbol{P}\boldsymbol{D}^3\boldsymbol{P}^{-1}\begin{pmatrix} 1000 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 350 \\ 312 \\ 338 \end{pmatrix}$, i.e. there will be 350 ants in A, 312 in B and 338 in C after 3 hours.

(c) As $n \to \infty$, $\boldsymbol{D}^n = \text{diag}(1^n, \left(\frac{2}{5}\right)^n, (-\frac{1}{10})^n) \to \text{diag}(1,0,0)$,

so $\boldsymbol{x}_n = \boldsymbol{A}^n\boldsymbol{x}_0 = \boldsymbol{P}\boldsymbol{D}^n\boldsymbol{P}^{-1}\boldsymbol{x}_0 \to \boldsymbol{P}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\boldsymbol{P}^{-1}\begin{pmatrix} 1000 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 333\frac{1}{3} \\ 333\frac{1}{3} \\ 333\frac{1}{3} \end{pmatrix}$.

Question 2: Solution

(a) $\begin{aligned} a_{n+1} &= 0.4a_n + 0.1b_n + 0.5c_n. \\ b_{n+1} &= 0.2a_n + 0.6b_n + 0.2c_n. \\ c_{n+1} &= 0.4a_n + 0.3b_n + 0.3c_n. \end{aligned}$ Thus, $\boldsymbol{A} = \begin{pmatrix} 0.4 & 0.1 & 0.5 \\ 0.2 & 0.6 & 0.2 \\ 0.4 & 0.3 & 0.3 \end{pmatrix}$.

(b) $\det(x\boldsymbol{I} - \boldsymbol{A}) = (x-1)(x-\frac{2}{5})(x+\frac{1}{10})$.

$E_1 = \text{span}\left\{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right\}$, $E_{\frac{2}{5}} = \text{span}\left\{\begin{pmatrix} 4 \\ -5 \\ 1 \end{pmatrix}\right\}$, $E_{-\frac{1}{10}} = \text{span}\left\{\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}\right\}$.

Let $\boldsymbol{P} = \begin{pmatrix} 1 & 4 & -1 \\ 1 & -5 & 0 \\ 1 & 1 & 1 \end{pmatrix}$, $\boldsymbol{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{2}{5} & 0 \\ 0 & 0 & -\frac{1}{10} \end{pmatrix}$. Then $\boldsymbol{P}^{-1} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{15} & -\frac{2}{15} & \frac{1}{15} \\ -\frac{2}{5} & -\frac{1}{5} & \frac{3}{5} \end{pmatrix}$ and $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$.

So $\boldsymbol{x}_3 = \boldsymbol{A}^3\boldsymbol{x}_0 = \boldsymbol{P}\boldsymbol{D}^3\boldsymbol{P}^{-1}\begin{pmatrix} 1000 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 350 \\ 312 \\ 338 \end{pmatrix}$, i.e. there will be 350 ants in A, 312 in B and 338 in C after 3 hours.

(c) As $n \to \infty$, $\boldsymbol{D}^n = \text{diag}(1^n, \left(\frac{2}{5}\right)^n, (-\frac{1}{10})^n) \to \text{diag}(1, 0, 0)$,

so $\boldsymbol{x}_n = \boldsymbol{A}^n\boldsymbol{x}_0 = \boldsymbol{P}\boldsymbol{D}^n\boldsymbol{P}^{-1}\boldsymbol{x}_0 \to \boldsymbol{P}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\boldsymbol{P}^{-1}\begin{pmatrix} 1000 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 333\frac{1}{3} \\ 333\frac{1}{3} \\ 333\frac{1}{3} \end{pmatrix}$.

Singular Value Decomposition (SVD)

Let $\boldsymbol{A}$ be an $m \times n$ matrix.

A **singular value decomposition of $\boldsymbol{A}$** is an expression

$$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$$

where $\boldsymbol{U}$ and $\boldsymbol{V}$ are orthogonal matrices of order m and n respectively, and $\boldsymbol{\Sigma} = (s_{ij})_{m\times n}$ is a 'diagonal' $m \times n$ matrix with decreasing non-negative 'diagonal' entries , i.e. $s_{ij} = 0$ whenever $i \neq j$ and $s_{11} \geq s_{22} \geq s_{33} \geq \cdots \geq 0$.

Conditions for SVD

Let $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ be a singular value decomposition of $\boldsymbol{A}$, with $\boldsymbol{U} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m \end{pmatrix}$, $\boldsymbol{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$, $\boldsymbol{\Sigma} = (s_{ij})_{m\times n}$. Then

$$\boldsymbol{A}^T\boldsymbol{A} = (\boldsymbol{U}^T\boldsymbol{\Sigma}\boldsymbol{V}^T)^T(\boldsymbol{U}^T\boldsymbol{\Sigma}\boldsymbol{V}^T) = \boldsymbol{V}\boldsymbol{\Sigma}^T\boldsymbol{U}^T\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T = \boldsymbol{V}(\boldsymbol{\Sigma}^T\boldsymbol{\Sigma})\boldsymbol{V}^T.$$

Since $\boldsymbol{\Sigma}^T\boldsymbol{\Sigma} = \operatorname{diag}(s_{11}^2, s_{22}^2, \ldots, s_{kk}^2, 0, \ldots, 0)$ where $k = \min\{m, n\}$ and $\boldsymbol{V}$ is orthogonal, this is an orthogonal diagonalization of $\boldsymbol{A}^T\boldsymbol{A}$.

Furthermore, $\boldsymbol{A}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{\Sigma}$, so that

$$\begin{pmatrix} \boldsymbol{A}\boldsymbol{v}_1 & \boldsymbol{A}\boldsymbol{v}_2 & \cdots & \boldsymbol{A}\boldsymbol{v}_n \end{pmatrix} = \begin{pmatrix} s_{11}\boldsymbol{u}_1 & s_{22}\boldsymbol{u}_2 & \cdots & s_{kk}\boldsymbol{u}_k & \boldsymbol{0} & \cdots & \boldsymbol{0} \end{pmatrix}.$$

Theorem

1. *The $\boldsymbol{v}_i$'s are orthonormal eigenvectors of $\boldsymbol{A}^T\boldsymbol{A}$.*
2. *The 'diagonal' entries s_{ii}'s of $\boldsymbol{\Sigma}$ are non-negative square-roots of the eigenvalues of $\boldsymbol{A}^T\boldsymbol{A}$, associated to the $\boldsymbol{v}_i$'s.*
3. *If $s_{ii} \neq 0$, then $\boldsymbol{u}_i = \frac{1}{s_{ii}}\boldsymbol{A}\boldsymbol{v}_i$.*

Conditions for SVD

Let $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ be a singular value decomposition of $\boldsymbol{A}$, with $\boldsymbol{U} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m \end{pmatrix}$, $\boldsymbol{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$, $\boldsymbol{\Sigma} = (s_{ij})_{m\times n}$. Then

$$\boldsymbol{A}^T\boldsymbol{A} = (\boldsymbol{U}^T\boldsymbol{\Sigma}\boldsymbol{V}^T)^T(\boldsymbol{U}^T\boldsymbol{\Sigma}\boldsymbol{V}^T) = \boldsymbol{V}\boldsymbol{\Sigma}^T\boldsymbol{U}^T\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T = \boldsymbol{V}(\boldsymbol{\Sigma}^T\boldsymbol{\Sigma})\boldsymbol{V}^T.$$

Since $\boldsymbol{\Sigma}^T\boldsymbol{\Sigma} = \operatorname{diag}(s_{11}^2, s_{22}^2, \ldots, s_{kk}^2, 0, \ldots, 0)$ where $k = \min\{m, n\}$ and $\boldsymbol{V}$ is orthogonal, this is an orthogonal diagonalization of $\boldsymbol{A}^T\boldsymbol{A}$.

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$$\begin{pmatrix} \boldsymbol{A}\boldsymbol{v}_1 & \boldsymbol{A}\boldsymbol{v}_2 & \cdots & \boldsymbol{A}\boldsymbol{v}_n \end{pmatrix} = \begin{pmatrix} s_{11}\boldsymbol{u}_1 & s_{22}\boldsymbol{u}_2 & \cdots & s_{kk}\boldsymbol{u}_k & \boldsymbol{0} & \cdots & \boldsymbol{0} \end{pmatrix}.$$

Theorem

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Conditions for SVD

Let $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ be a singular value decomposition of $\boldsymbol{A}$, with $\boldsymbol{U} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m \end{pmatrix}$, $\boldsymbol{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$, $\boldsymbol{\Sigma} = (s_{ij})_{m\times n}$. Then

$$\boldsymbol{A}^T\boldsymbol{A} = (\boldsymbol{U}^T\boldsymbol{\Sigma}\boldsymbol{V}^T)^T(\boldsymbol{U}^T\boldsymbol{\Sigma}\boldsymbol{V}^T) = \boldsymbol{V}\boldsymbol{\Sigma}^T\boldsymbol{U}^T\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T = \boldsymbol{V}(\boldsymbol{\Sigma}^T\boldsymbol{\Sigma})\boldsymbol{V}^T.$$

Since $\boldsymbol{\Sigma}^T\boldsymbol{\Sigma} = \operatorname{diag}(s_{11}^2, s_{22}^2, \ldots, s_{kk}^2, 0, \ldots, 0)$ where $k = \min\{m, n\}$ and $\boldsymbol{V}$ is orthogonal, this is an orthogonal diagonalization of $\boldsymbol{A}^T\boldsymbol{A}$.

Furthermore, $\boldsymbol{A}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{\Sigma}$, so that

$$\begin{pmatrix} \boldsymbol{A}\boldsymbol{v}_1 & \boldsymbol{A}\boldsymbol{v}_2 & \cdots & \boldsymbol{A}\boldsymbol{v}_n \end{pmatrix} = \begin{pmatrix} s_{11}\boldsymbol{u}_1 & s_{22}\boldsymbol{u}_2 & \cdots & s_{kk}\boldsymbol{u}_k & \boldsymbol{0} & \cdots & \boldsymbol{0} \end{pmatrix}.$$

Theorem

1. *The $\boldsymbol{v}_i$'s are orthonormal eigenvectors of $\boldsymbol{A}^T\boldsymbol{A}$.*
2. *The 'diagonal' entries s_{ii}'s of $\boldsymbol{\Sigma}$ are non-negative square-roots of the eigenvalues of $\boldsymbol{A}^T\boldsymbol{A}$, associated to the $\boldsymbol{v}_i$'s.*
3. *If $s_{ii} \neq 0$, then $\boldsymbol{u}_i = \frac{1}{s_{ii}}\boldsymbol{A}\boldsymbol{v}_i$.*

Conditions for SVD

Let $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ be a singular value decomposition of $\boldsymbol{A}$, with $\boldsymbol{U} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m \end{pmatrix}$, $\boldsymbol{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$, $\boldsymbol{\Sigma} = (s_{ij})_{m\times n}$. Then

$$\boldsymbol{A}^T\boldsymbol{A} = (\boldsymbol{U}^T\boldsymbol{\Sigma}\boldsymbol{V}^T)^T(\boldsymbol{U}^T\boldsymbol{\Sigma}\boldsymbol{V}^T) = \boldsymbol{V}\boldsymbol{\Sigma}^T\boldsymbol{U}^T\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T = \boldsymbol{V}(\boldsymbol{\Sigma}^T\boldsymbol{\Sigma})\boldsymbol{V}^T.$$

Since $\boldsymbol{\Sigma}^T\boldsymbol{\Sigma} = \operatorname{diag}(s_{11}^2, s_{22}^2, \ldots, s_{kk}^2, 0, \ldots, 0)$ where $k = \min\{m, n\}$ and $\boldsymbol{V}$ is orthogonal, this is an orthogonal diagonalization of $\boldsymbol{A}^T\boldsymbol{A}$.

Furthermore, $\boldsymbol{A}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{\Sigma}$, so that

$$\begin{pmatrix} \boldsymbol{A}\boldsymbol{v}_1 & \boldsymbol{A}\boldsymbol{v}_2 & \cdots & \boldsymbol{A}\boldsymbol{v}_n \end{pmatrix} = \begin{pmatrix} s_{11}\boldsymbol{u}_1 & s_{22}\boldsymbol{u}_2 & \cdots & s_{kk}\boldsymbol{u}_k & \boldsymbol{0} & \cdots & \boldsymbol{0} \end{pmatrix}.$$

Theorem

1. *The $\boldsymbol{v}_i$'s are orthonormal eigenvectors of $\boldsymbol{A}^T\boldsymbol{A}$.*
2. *The 'diagonal' entries s_{ii}'s of $\boldsymbol{\Sigma}$ are non-negative square-roots of the eigenvalues of $\boldsymbol{A}^T\boldsymbol{A}$, associated to the $\boldsymbol{v}_i$'s.*
3. *If $s_{ii} \neq 0$, then $\boldsymbol{u}_i = \frac{1}{s_{ii}}\boldsymbol{A}\boldsymbol{v}_i$.*

Conditions for SVD

Let $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ be a singular value decomposition of $\boldsymbol{A}$, with $\boldsymbol{U} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m \end{pmatrix}$, $\boldsymbol{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$, $\boldsymbol{\Sigma} = (s_{ij})_{m\times n}$. Then

$$\boldsymbol{A}^T\boldsymbol{A} = (\boldsymbol{U}^T\boldsymbol{\Sigma}\boldsymbol{V}^T)^T(\boldsymbol{U}^T\boldsymbol{\Sigma}\boldsymbol{V}^T) = \boldsymbol{V}\boldsymbol{\Sigma}^T\boldsymbol{U}^T\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T = \boldsymbol{V}(\boldsymbol{\Sigma}^T\boldsymbol{\Sigma})\boldsymbol{V}^T.$$

Since $\boldsymbol{\Sigma}^T\boldsymbol{\Sigma} = \operatorname{diag}(s_{11}^2, s_{22}^2, \ldots, s_{kk}^2, 0, \ldots, 0)$ where $k = \min\{m, n\}$ and $\boldsymbol{V}$ is orthogonal, this is an orthogonal diagonalization of $\boldsymbol{A}^T\boldsymbol{A}$.

Furthermore, $\boldsymbol{A}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{\Sigma}$, so that

$$\begin{pmatrix} \boldsymbol{A}\boldsymbol{v}_1 & \boldsymbol{A}\boldsymbol{v}_2 & \cdots & \boldsymbol{A}\boldsymbol{v}_n \end{pmatrix} = \begin{pmatrix} s_{11}\boldsymbol{u}_1 & s_{22}\boldsymbol{u}_2 & \cdots & s_{kk}\boldsymbol{u}_k & \boldsymbol{0} & \cdots & \boldsymbol{0} \end{pmatrix}.$$

Theorem

1. *The $\boldsymbol{v}_i$'s are orthonormal eigenvectors of $\boldsymbol{A}^T\boldsymbol{A}$.*
2. *The 'diagonal' entries s_{ii}'s of Σ are non-negative square-roots of the eigenvalues of $\boldsymbol{A}^T\boldsymbol{A}$, associated to the $\boldsymbol{v}_i$'s.*
3. *If $s_{ii} \neq 0$, then $\boldsymbol{u}_i = \frac{1}{s_{ii}}\boldsymbol{A}\boldsymbol{v}_i$.*

Conditions for SVD

Let $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ be a singular value decomposition of $\boldsymbol{A}$, with $\boldsymbol{U} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m \end{pmatrix}$, $\boldsymbol{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$, $\boldsymbol{\Sigma} = (s_{ij})_{m\times n}$. Then

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Since $\boldsymbol{\Sigma}^T\boldsymbol{\Sigma} = \operatorname{diag}(s_{11}^2, s_{22}^2, \ldots, s_{kk}^2, 0, \ldots, 0)$ where $k = \min\{m, n\}$ and $\boldsymbol{V}$ is orthogonal, this is an orthogonal diagonalization of $\boldsymbol{A}^T\boldsymbol{A}$.

Furthermore, $\boldsymbol{A}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{\Sigma}$, so that

$$\begin{pmatrix} \boldsymbol{A}\boldsymbol{v}_1 & \boldsymbol{A}\boldsymbol{v}_2 & \cdots & \boldsymbol{A}\boldsymbol{v}_n \end{pmatrix} = \begin{pmatrix} s_{11}\boldsymbol{u}_1 & s_{22}\boldsymbol{u}_2 & \cdots & s_{kk}\boldsymbol{u}_k & \boldsymbol{0} & \cdots & \boldsymbol{0} \end{pmatrix}.$$

Theorem

1. *The $\boldsymbol{v}_i$'s are orthonormal eigenvectors of $\boldsymbol{A}^T\boldsymbol{A}$.*
2. *The 'diagonal' entries s_{ii}'s of $\boldsymbol{\Sigma}$ are non-negative square-roots of the eigenvalues of $\boldsymbol{A}^T\boldsymbol{A}$, associated to the $\boldsymbol{v}_i$'s.*
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Conditions for SVD

Let $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ be a singular value decomposition of $\boldsymbol{A}$, with $\boldsymbol{U} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m \end{pmatrix}$, $\boldsymbol{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$, $\boldsymbol{\Sigma} = (s_{ij})_{m\times n}$. Then

$$\boldsymbol{A}^T\boldsymbol{A} = (\boldsymbol{U}^T\boldsymbol{\Sigma}\boldsymbol{V}^T)^T(\boldsymbol{U}^T\boldsymbol{\Sigma}\boldsymbol{V}^T) = \boldsymbol{V}\boldsymbol{\Sigma}^T\boldsymbol{U}^T\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T = \boldsymbol{V}(\boldsymbol{\Sigma}^T\boldsymbol{\Sigma})\boldsymbol{V}^T.$$

Since $\boldsymbol{\Sigma}^T\boldsymbol{\Sigma} = \operatorname{diag}(s_{11}^2, s_{22}^2, \ldots, s_{kk}^2, 0, \ldots, 0)$ where $k = \min\{m, n\}$ and $\boldsymbol{V}$ is orthogonal, this is an orthogonal diagonalization of $\boldsymbol{A}^T\boldsymbol{A}$.

Furthermore, $\boldsymbol{A}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{\Sigma}$, so that

$$\begin{pmatrix} \boldsymbol{A}\boldsymbol{v}_1 & \boldsymbol{A}\boldsymbol{v}_2 & \cdots & \boldsymbol{A}\boldsymbol{v}_n \end{pmatrix} = \begin{pmatrix} s_{11}\boldsymbol{u}_1 & s_{22}\boldsymbol{u}_2 & \cdots & s_{kk}\boldsymbol{u}_k & \boldsymbol{0} & \cdots & \boldsymbol{0} \end{pmatrix}.$$

Theorem

1. *The $\boldsymbol{v}_i$'s are orthonormal eigenvectors of $\boldsymbol{A}^T\boldsymbol{A}$.*
2. *The 'diagonal' entries s_{ii}'s of $\boldsymbol{\Sigma}$ are non-negative square-roots of the eigenvalues of $\boldsymbol{A}^T\boldsymbol{A}$, associated to the $\boldsymbol{v}_i$'s.*
3. *If $s_{ii} \neq 0$, then $\boldsymbol{u}_i = \frac{1}{s_{ii}}\boldsymbol{A}\boldsymbol{v}_i$.*

Algorithm for SVD

Let $\boldsymbol{A}$ be an $m \times n$ matrix.

1. Orthogonally diagonalize the symmetric matrix $\boldsymbol{A}^T\boldsymbol{A}$; i.e. find orthogonal matrix $\boldsymbol{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$ and diagonal matrix $\boldsymbol{D} = \operatorname{diag}(d_1, d_2, \ldots, d_n)$ such that $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{V}\boldsymbol{D}\boldsymbol{V}^T$.
   We order the columns of $\boldsymbol{V}$ so that $d_1 \geq d_2 \geq \cdots \geq d_n$ (≥ 0).
2. Let $r = \max\{i \mid d_i > 0\}$. For each $1 \leq i \leq r$, let $\boldsymbol{u}_i = \frac{1}{\sqrt{d_i}}\boldsymbol{A}\boldsymbol{v}_i$. Then $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\}$ is orthonormal.
3. Extend $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\}$ to an orthonormal basis $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r, \boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m\}$ for $\mathbb{R}^m$ by Gram-Schmidt process.
4. Let $\boldsymbol{U} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m \end{pmatrix}$ and $\boldsymbol{\Sigma} = (s_{ij})_{m\times n}$ where

$$s_{ij} = \begin{cases} \sqrt{d_i}, & \text{if } i = j \leq r; \\ 0, & \text{otherwise.} \end{cases}$$

Then $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ is a singular value decomposition for $\boldsymbol{A}$.

Algorithm for SVD

Let $\boldsymbol{A}$ be an $m \times n$ matrix.

1. Orthogonally diagonalize the symmetric matrix $\boldsymbol{A}^T\boldsymbol{A}$; i.e. find orthogonal matrix $\boldsymbol{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$ and diagonal matrix $\boldsymbol{D} = \operatorname{diag}(d_1, d_2, \ldots, d_n)$ such that $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{V}\boldsymbol{D}\boldsymbol{V}^T$.
   We order the columns of $\boldsymbol{V}$ so that $d_1 \geq d_2 \geq \cdots \geq d_n$ (≥ 0).
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3. Extend $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\}$ to an orthonormal basis $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r, \boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m\}$ for $\mathbb{R}^m$ by Gram-Schmidt process.
4. Let $\boldsymbol{U} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m \end{pmatrix}$ and $\boldsymbol{\Sigma} = (s_{ij})_{m\times n}$ where

$$s_{ij} = \begin{cases} \sqrt{d_i}, & \text{if } i = j \leq r; \\ 0, & \text{otherwise.} \end{cases}$$

Then $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ is a singular value decomposition for $\boldsymbol{A}$.

Algorithm for SVD

Let $\boldsymbol{A}$ be an $m \times n$ matrix.

1. Orthogonally diagonalize the symmetric matrix $\boldsymbol{A}^T\boldsymbol{A}$; i.e. find orthogonal matrix $\boldsymbol{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$ and diagonal matrix $\boldsymbol{D} = \operatorname{diag}(d_1, d_2, \ldots, d_n)$ such that $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{V}\boldsymbol{D}\boldsymbol{V}^T$.
   We order the columns of $\boldsymbol{V}$ so that $d_1 \geq d_2 \geq \cdots \geq d_n$ (≥ 0).
2. Let $r = \max\{i \mid d_i > 0\}$. For each $1 \leq i \leq r$, let $\boldsymbol{u}_i = \frac{1}{\sqrt{d_i}}\boldsymbol{A}\boldsymbol{v}_i$. Then $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\}$ is orthonormal.
3. Extend $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\}$ to an orthonormal basis $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r, \boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m\}$ for $\mathbb{R}^m$ by Gram-Schmidt process.
4. Let $\boldsymbol{U} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m \end{pmatrix}$ and $\boldsymbol{\Sigma} = (s_{ij})_{m\times n}$ where

$$s_{ij} = \begin{cases} \sqrt{d_i}, & \text{if } i = j \leq r; \\ 0, & \text{otherwise.} \end{cases}$$

Then $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ is a singular value decomposition for $\boldsymbol{A}$.

Algorithm for SVD

Let $\boldsymbol{A}$ be an $m \times n$ matrix.

1. Orthogonally diagonalize the symmetric matrix $\boldsymbol{A}^T\boldsymbol{A}$; i.e. find orthogonal matrix $\boldsymbol{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$ and diagonal matrix $\boldsymbol{D} = \operatorname{diag}(d_1, d_2, \ldots, d_n)$ such that $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{V}\boldsymbol{D}\boldsymbol{V}^T$.
   We order the columns of $\boldsymbol{V}$ so that $d_1 \geq d_2 \geq \cdots \geq d_n$ (≥ 0).
2. Let $r = \max\{i \mid d_i > 0\}$. For each $1 \leq i \leq r$, let $\boldsymbol{u}_i = \frac{1}{\sqrt{d_i}}\boldsymbol{A}\boldsymbol{v}_i$. Then $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\}$ is orthonormal.
3. Extend $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\}$ to an orthonormal basis $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r, \boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m\}$ for $\mathbb{R}^m$ by Gram-Schmidt process.
4. Let $\boldsymbol{U} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m \end{pmatrix}$ and $\boldsymbol{\Sigma} = (s_{ij})_{m\times n}$ where

$$s_{ij} = \begin{cases} \sqrt{d_i}, & \text{if } i = j \leq r; \\ 0, & \text{otherwise.} \end{cases}$$

Then $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ is a singular value decomposition for $\boldsymbol{A}$.

Algorithm for SVD

Let $\boldsymbol{A}$ be an $m \times n$ matrix.

1. Orthogonally diagonalize the symmetric matrix $\boldsymbol{A}^T\boldsymbol{A}$; i.e. find orthogonal matrix $\boldsymbol{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$ and diagonal matrix $\boldsymbol{D} = \operatorname{diag}(d_1, d_2, \ldots, d_n)$ such that $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{V}\boldsymbol{D}\boldsymbol{V}^T$.
   We order the columns of $\boldsymbol{V}$ so that $d_1 \geq d_2 \geq \cdots \geq d_n$ (≥ 0).
2. Let $r = \max\{i \mid d_i > 0\}$. For each $1 \leq i \leq r$, let $\boldsymbol{u}_i = \frac{1}{\sqrt{d_i}}\boldsymbol{A}\boldsymbol{v}_i$. Then $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\}$ is orthonormal.
3. Extend $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\}$ to an orthonormal basis $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r, \boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m\}$ for $\mathbb{R}^m$ by Gram-Schmidt process.
4. Let $\boldsymbol{U} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m \end{pmatrix}$ and $\boldsymbol{\Sigma} = (s_{ij})_{m\times n}$ where

$$s_{ij} = \begin{cases} \sqrt{d_i}, & \text{if } i = j \leq r; \\ 0, & \text{otherwise.} \end{cases}$$

Then $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ is a singular value decomposition for $\boldsymbol{A}$.

Question 3

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Find a singular value decomposition of $\boldsymbol{A} = \begin{pmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{pmatrix}$.

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Question 3: Solution

$\boldsymbol{A}^T\boldsymbol{A} = \begin{pmatrix} 17 & 8 \\ 8 & 17 \end{pmatrix}$. $\det(xI - \boldsymbol{A}^T\boldsymbol{A}) = (x-17)^2 - 64 = (x-25)(x-9)$.

$E_{25} = \operatorname{span}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}$, $E_9 = \operatorname{span}\left\{\begin{pmatrix} -1 \\ 1 \end{pmatrix}\right\}$, so $\boldsymbol{V} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$ and $\boldsymbol{\Sigma} = \begin{pmatrix} 5 & 0 \\ 0 & 3 \\ 0 & 0 \end{pmatrix}$.

$\boldsymbol{u}_1 = \frac{1}{5}\boldsymbol{A}\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}$, $\boldsymbol{u}_2 = \frac{1}{3}\boldsymbol{A}\begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} -\frac{1}{3\sqrt{2}} \\ \frac{1}{3\sqrt{2}} \\ -\frac{4}{3\sqrt{2}} \end{pmatrix}$.

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Thus $\boldsymbol{U} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{3\sqrt{2}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} & -\frac{2}{3} \\ 0 & -\frac{4}{3\sqrt{2}} & -\frac{1}{3} \end{pmatrix}$, and $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$.

Question 3: Solution

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Thus $\boldsymbol{U} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{3\sqrt{2}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} & -\frac{2}{3} \\ 0 & -\frac{4}{3\sqrt{2}} & -\frac{1}{3} \end{pmatrix}$, and $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$.

Question 3: Solution

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$\boldsymbol{u}_1 = \frac{1}{5}\boldsymbol{A}\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}$, $\boldsymbol{u}_2 = \frac{1}{3}\boldsymbol{A}\begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} -\frac{1}{3\sqrt{2}} \\ \frac{1}{3\sqrt{2}} \\ -\frac{4}{3\sqrt{2}} \end{pmatrix}$.

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Thus $\boldsymbol{U} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{3\sqrt{2}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} & -\frac{2}{3} \\ 0 & -\frac{4}{3\sqrt{2}} & -\frac{1}{3} \end{pmatrix}$, and $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$.

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Thus $\boldsymbol{U} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{3\sqrt{2}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} & -\frac{2}{3} \\ 0 & -\frac{4}{3\sqrt{2}} & -\frac{1}{3} \end{pmatrix}$, and $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$.

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Thus $\boldsymbol{U} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{3\sqrt{2}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} & -\frac{2}{3} \\ 0 & -\frac{4}{3\sqrt{2}} & -\frac{1}{3} \end{pmatrix}$, and $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$.

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Thus $\boldsymbol{U} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{3\sqrt{2}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} & -\frac{2}{3} \\ 0 & -\frac{4}{3\sqrt{2}} & -\frac{1}{3} \end{pmatrix}$, and $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$.

Next Lesson ...

We will explore **Section 7.1**.
Please watch/review the video clips for this section before then.